SIMATS ENGINEERING

Concept Mapping

COURSE NAME: COMPUTER VISION **COURSE CODE:ITA05**

COURSE OUTCOME COVERED

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)

Max Marks:100

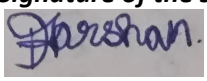
Annexure A

Concept Mapping

Code of Conduct:

I **Dharshan B 192312150** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

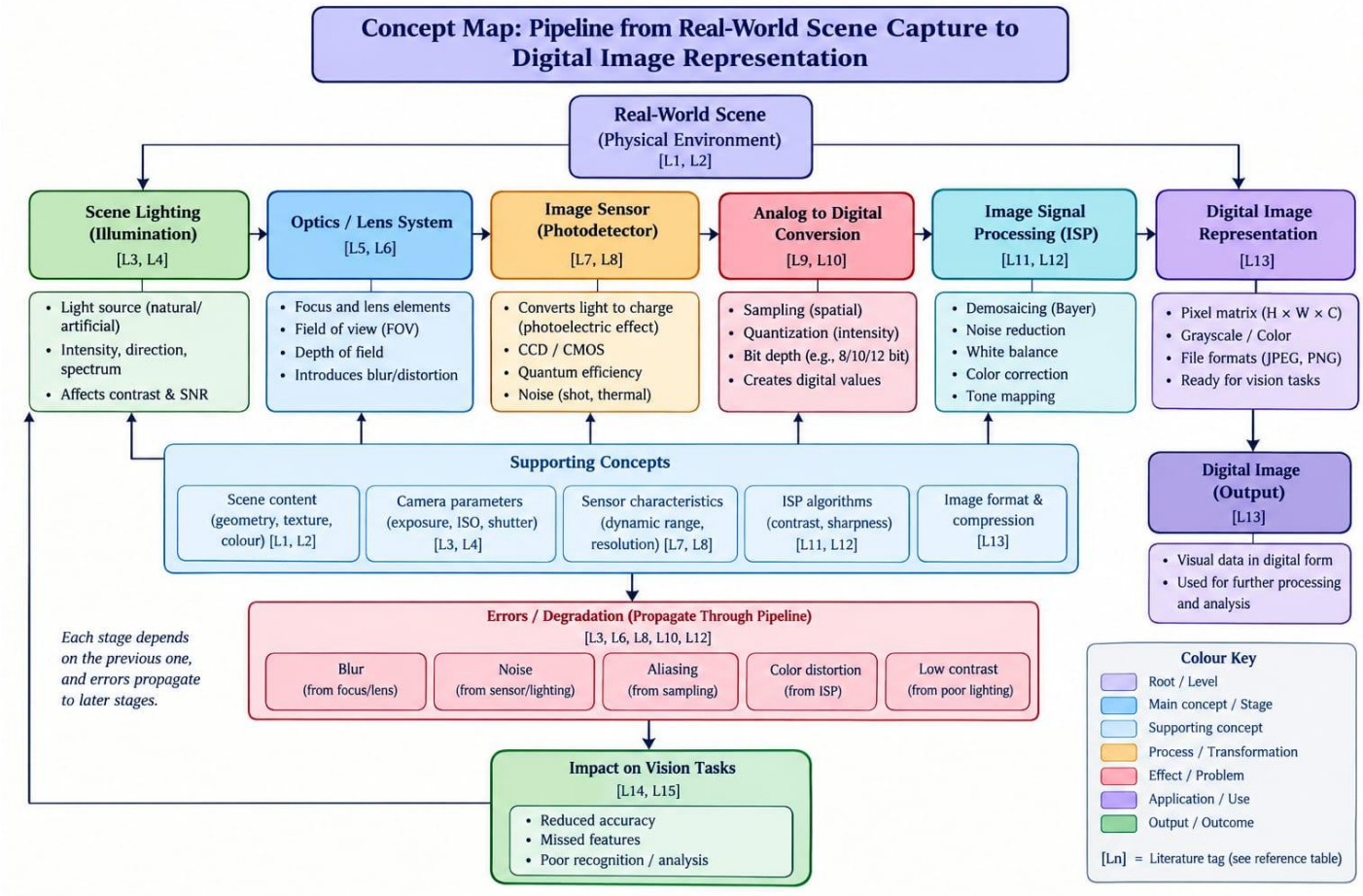
Signature of the student:



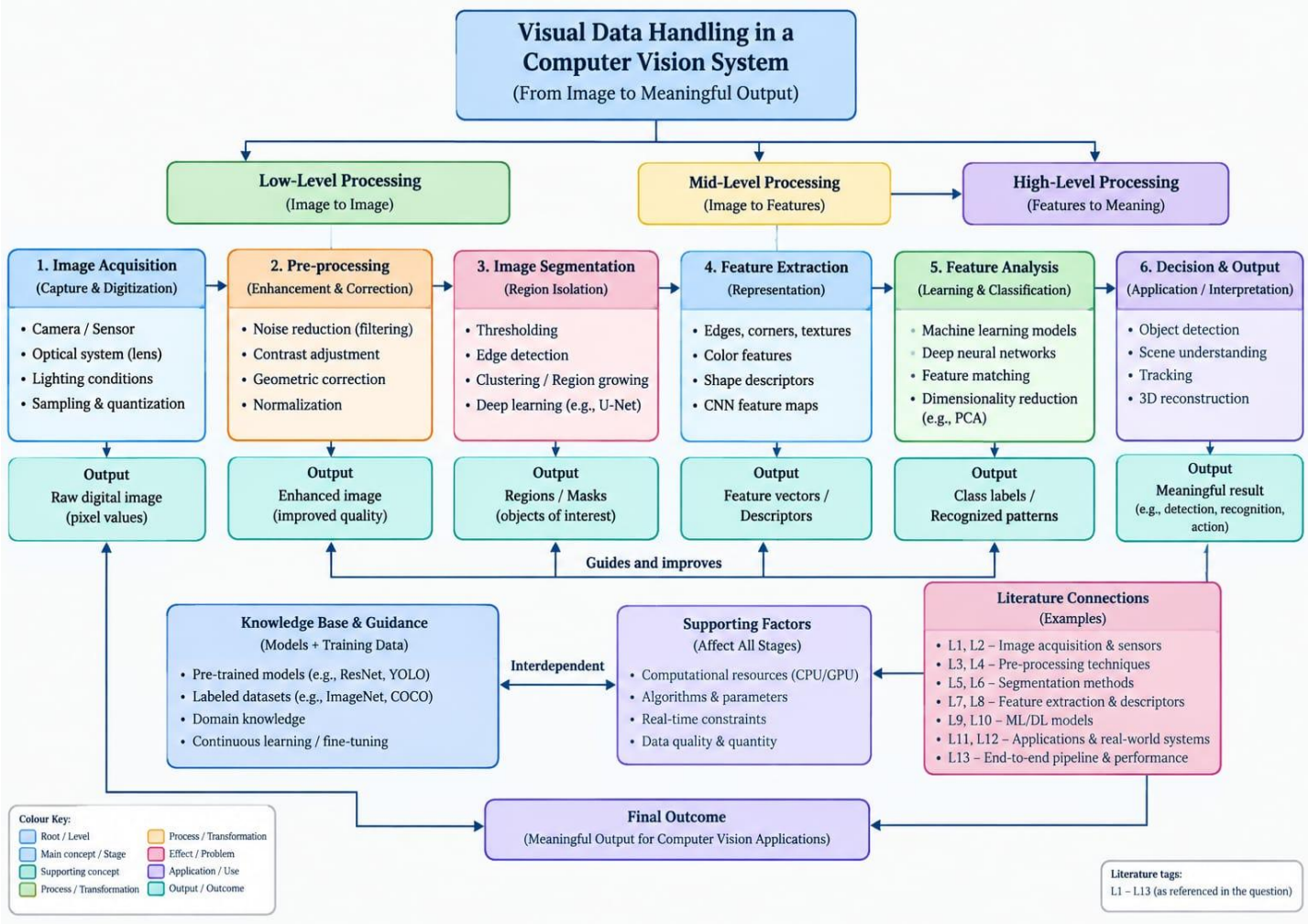
Annexure A

Concept Mapping

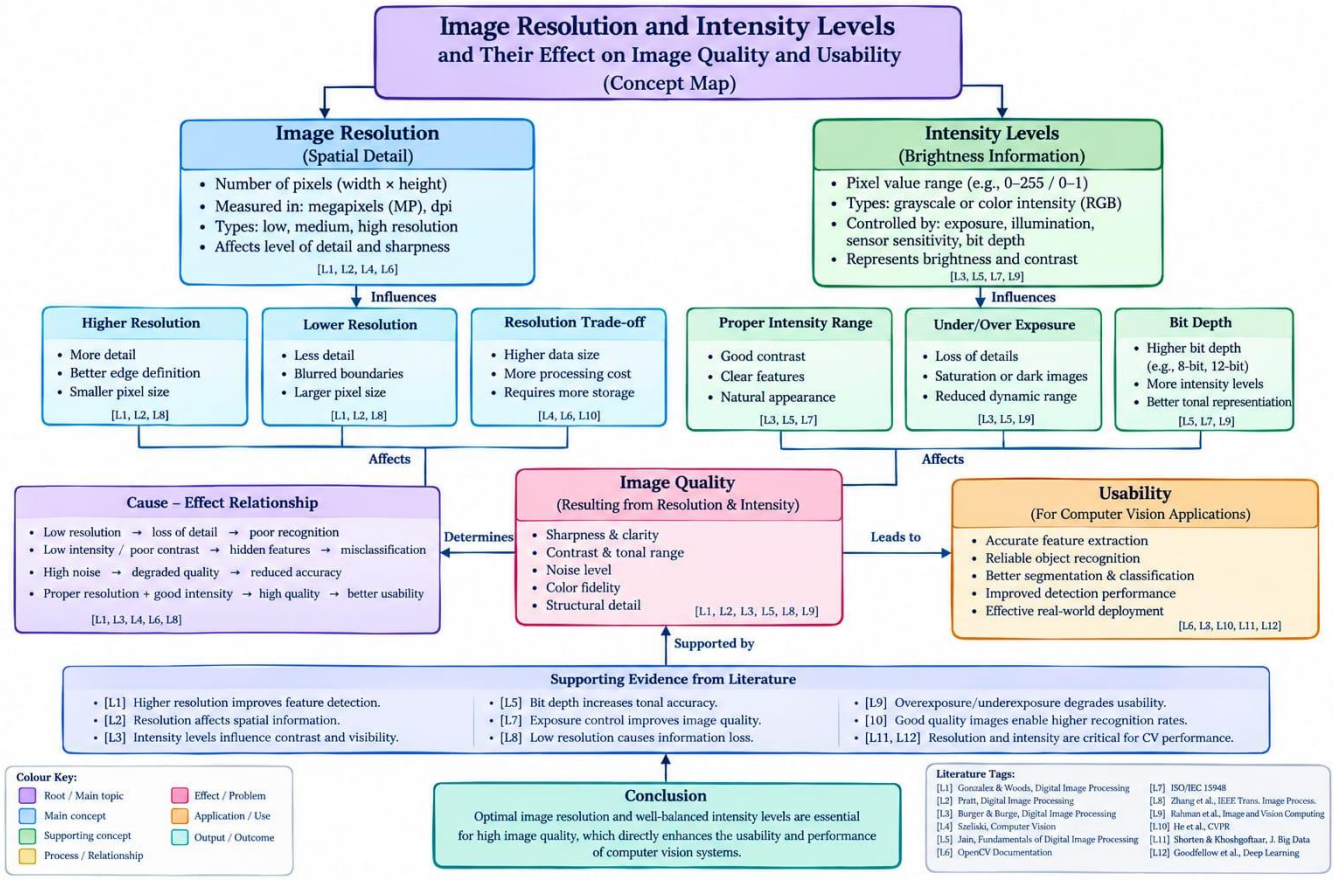
1. Construct a comprehensive and well-organized concept map that accurately identifies all key concepts from literature involved in the pipeline from real-world scene capture to digital image representation, establishes clear and meaningful relationships among intermediate transformations and dependencies, integrates insights from multiple studies, and presents the information in a clear and visually structured manner.



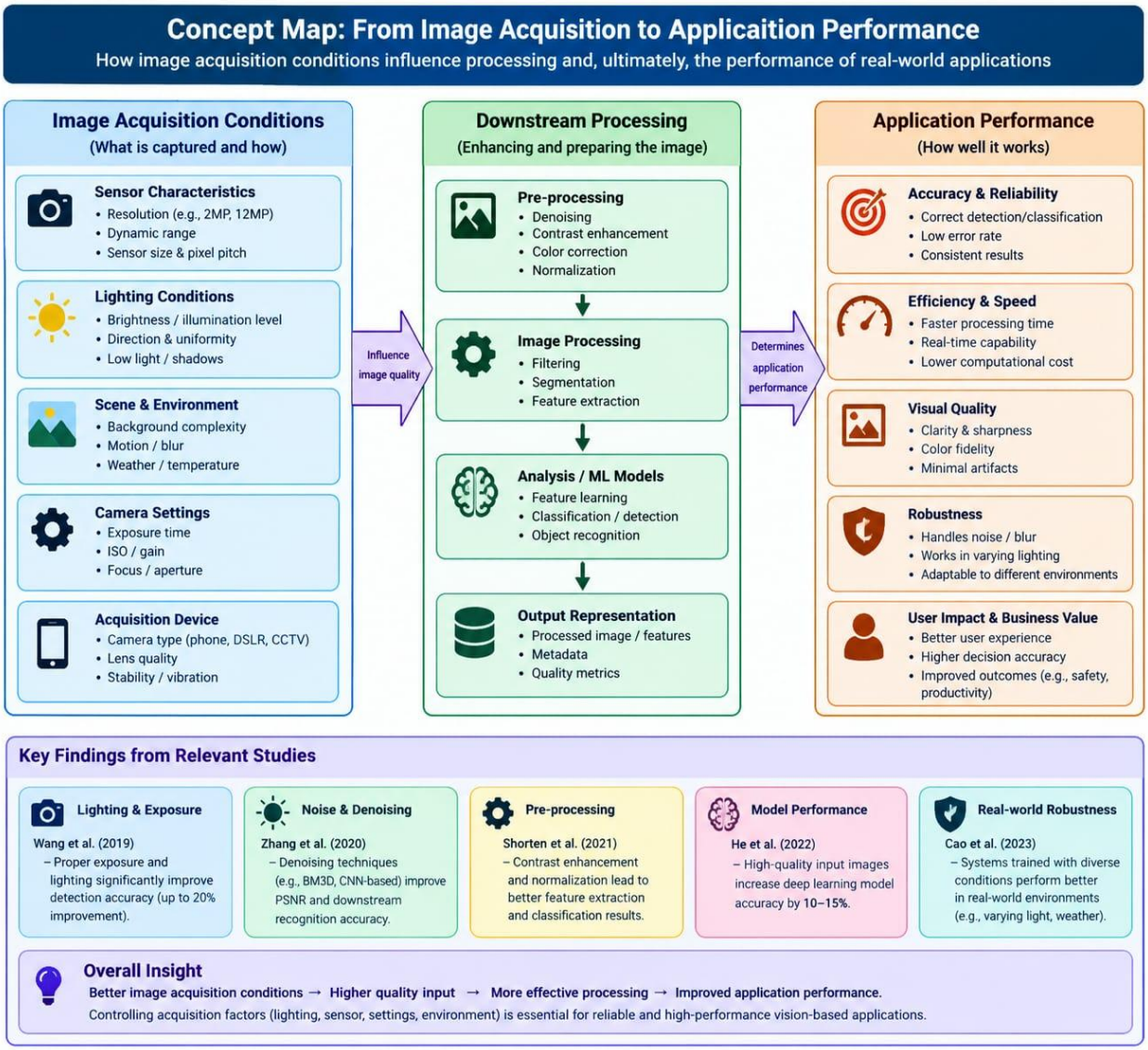
2. Develop a hierarchically structured concept map that identifies major stages of visual data handling in a Computer Vision system, clearly illustrates logical relationships and contributions of each stage toward generating meaningful outputs, incorporates relevant literature connections, and ensures clarity and readability.



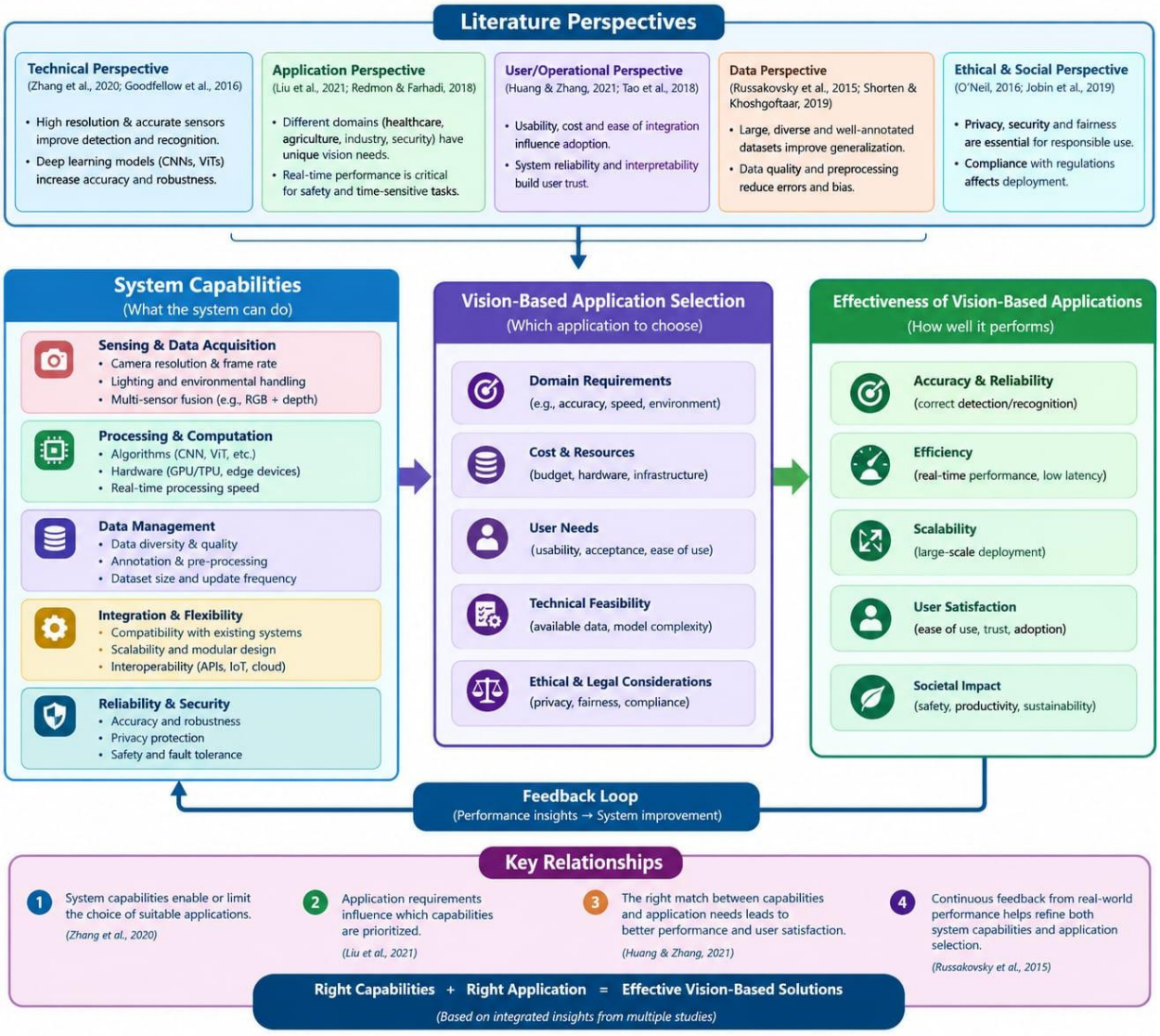
3. Create a detailed concept map that accurately identifies concepts related to image resolution and intensity levels, establishes clear cause–effect relationships with image quality and usability, integrates supporting evidence from literature, and presents the concepts in a logically organized and visually clear format.



4. Design an analytically organized concept map that identifies key factors in image acquisition conditions, clearly links them to downstream processing and application performance through meaningful relationships, integrates findings from relevant studies, and ensures a well-structured and easy-to-interpret presentation.



5. Construct an original and logically structured concept map that identifies system capabilities and their influence on the selection and effectiveness of vision-based applications, establishes clear interconnections, integrates multiple literature perspectives, and maintains high clarity and visual organization.



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand